

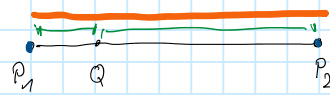
Ü-Shading-Muster

1) Gauß- Shading:

geg: $P_1(4|11)$ $P_2(10|5)$ $Q(5|10)$

$$I_{1,2} = \begin{Bmatrix} 0 \\ 0,25 \\ 0 \end{Bmatrix}$$

Variante 1:



$$I_Q = \frac{y_Q - y_{P_2}}{y_{P_1} - y_{P_2}} \cdot I_{P_1} + \frac{y_{P_1} - y_Q}{y_{P_1} - y_{P_2}} \cdot I_{P_2}$$

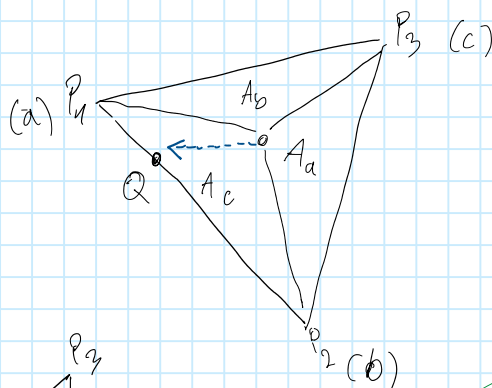
I antiproportional zur Strecke

$$I_Q = \frac{10 - 5}{11 - 5} \cdot I_{P_1} + \frac{11 - 10}{11 - 5} \cdot I_{P_2}$$

$$= \frac{5}{6} \cdot I_{P_1} + \frac{1}{6} \cdot I_{P_2}$$

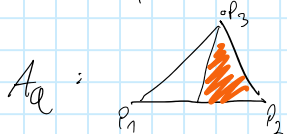
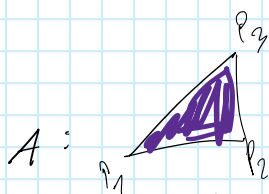
$$I_Q = \frac{5}{6} \cdot \begin{Bmatrix} 0 \\ 0,25 \\ 0 \end{Bmatrix} + \frac{1}{6} \begin{Bmatrix} 0 \\ 0,25 \\ 0 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 5/24 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 1/24 \\ 0 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 6/24 \\ 0 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0,25 \\ 0 \end{Bmatrix}$$

Variante 2:



Flächeninhalt eines Dreiecks:

$$A = \frac{1}{2} \cdot g \cdot h$$

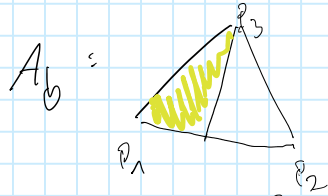


abgelesen in Skizze:

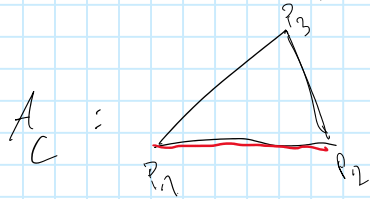
$$A = \frac{1}{2} \cdot 6\sqrt{2} \cdot 5\sqrt{2} = 30$$

Kästchen-Diagonale: $\sqrt{2}$
(Hervorheben mit Pythagoras)

$$\frac{1}{2} \cdot 5\sqrt{2} \cdot 5\sqrt{2} = \frac{50}{2} = 25$$



$$A - A_a = 30 - 25 = 5$$



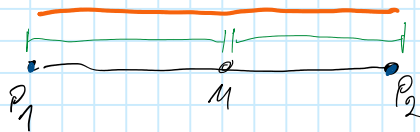
$$A_c = 0$$

$$\rho(\alpha, \beta, \gamma) = \alpha a + \beta b + \gamma c \quad \left| \quad \begin{aligned} \alpha &= \frac{A_a}{A} = \frac{25}{30} = \frac{5}{6} \\ \beta &= \frac{A_b}{A} = \frac{5}{30} = \frac{1}{6} \\ \gamma &= \frac{A_c}{A} = 0 \end{aligned} \right.$$

$$= \frac{5}{6} \cdot I_{p_1} + \frac{1}{6} \cdot I_{p_2} + 0 \cdot I_{p_3}$$

2) Phong-Shading:

Variante 1:



$$\vec{N}_m = \frac{y_m - y_{p_2}}{y_{p_1} - y_{p_2}} \cdot \vec{N}_1 + \frac{y_{p_1} - y_m}{y_{p_1} - y_{p_2}} \cdot \vec{N}_2$$

$$\vec{N}_m = \frac{8-5}{11-5} \vec{N}_1 + \frac{11-8}{11-5} \vec{N}_2 = \frac{1}{2} \vec{N}_1 + \frac{1}{2} \vec{N}_2$$

$$\vec{N}_m = \frac{1}{2} \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix} \leftarrow \text{noch nicht normiert!}$$

abgelesen
aus Skizze

$$\bar{N}_m = \frac{\begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix}}{\sqrt{\left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2}} = \begin{pmatrix} -1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

$$\begin{aligned} \bar{L}_m &= \boxed{\frac{1}{2}} \cdot \bar{L}_1 + \boxed{\frac{1}{2}} \cdot \bar{L}_2 \\ &= \frac{1}{2} \begin{pmatrix} 0 \\ -1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix} \end{aligned}$$

← nicht normalisiert!

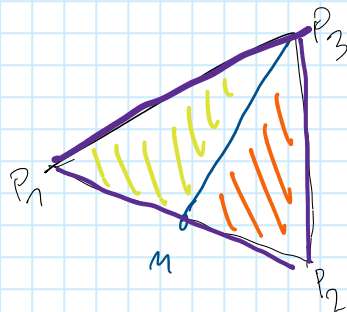
← Vorzeichen bleiben

← abgelesen aus Skizze

$$\bar{L}_m = \bar{N}_m = \begin{pmatrix} -1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \quad \bar{N}_m \cdot \bar{L}_m = 1$$

$$\begin{aligned} I_m &= I_a \cdot k_a + I_L \cdot k_d \cdot (\bar{N}_m \cdot \bar{L}_m) + \underbrace{I_L \cdot k_s}_{\text{enthält da } k_s = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}} (\bar{v} \cdot \bar{R})^n \\ &= \begin{Bmatrix} 0,5 \\ 0,5 \\ 0,5 \end{Bmatrix} \cdot \begin{Bmatrix} 0 \\ 0,5 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0,5 \\ 0,5 \\ 0,5 \end{Bmatrix} \cdot \begin{Bmatrix} 0,5 \\ 0 \\ 0 \end{Bmatrix} \cdot 1 \\ &= \begin{Bmatrix} 0,25 \\ 0,25 \\ 0 \end{Bmatrix} \end{aligned}$$

Variante 2:



Dreieck wird genau halbiert

$$A_b = A_a = \frac{1}{2} A$$